

CISCO SECURE WORKLOAD

NIST 800-53 Rev 5 Compliance Framework

Security & Privacy Controls — CSW-Relevant Families

Prepared for: [Customer Name]

Prepared by: Cisco Systems, Inc.

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Regulatory Standard: NIST SP 800-53 Rev 5 (AC, AU, CM, IR, RA, SC, SI Families)

Environment: Hybrid (On-Premises + Cloud)

Deployment Stage: Fresh Installation

How to read the Coverage column

Direct means Secure Workload produces the primary evidence artifact for that row — segmentation, policy, flow and process telemetry, inventory, vulnerability and reachability, or forensic reconstruction. Direct is still evidence, not an attestation.

Supporting means Secure Workload contributes an input that feeds or complements another control, tool, or process — detection input only, reconciled against an external register, or one log source among several.

Evidence Required means Secure Workload has no record for that row. The customer supplies the contract, the analysis, or the HR / physical-security evidence from another control.

Every report in this set now uses the same three coverage words — Direct, Supporting, and Evidence Required. Coverage describes what evidence Secure Workload produces; it never means the control is passed.

Executive Summary

NIST SP 800-53 Rev 5 provides the comprehensive security control catalog for federal information systems and is widely adopted in regulated commercial environments. This report maps Cisco Secure Workload (CSW) capabilities to the seven control families most directly addressed: Access Control (AC), Audit & Accountability (AU), Configuration Management (CM), Incident Response (IR), Risk Assessment (RA), System & Communications Protection (SC), and System & Information Integrity (SI).

CSW achieves Direct coverage across all 18 mapped controls through workload-level policy enforcement, continuous telemetry, vulnerability management, and forensic investigation capabilities.

Control Coverage by Family

Family	Control	Name	NIST Requirement	CSW Implementation	Coverage
AC	AC-3	Access Enforcement	Enforce approved authorizations for logical access	Micro-segmentation enforces allowlist policies at workload level; default deny	Direct
AC	AC-4	Information Flow Enforcement	Control information flows between systems	ADM maps all flows; policy blocks unauthorized data paths across all environments	Direct
AC	AC-6	Least Privilege	Employ least privilege for access to system resources	Workload-scoped policies grant minimum required network access per service	Direct
AC	AC-17	Remote Access	Establish usage restrictions for remote access	Jump-host enforcement; remote admin paths explicitly defined and logged	Direct
AU	AU-2	Event Logging	Identify the types of events the system is capable of logging	Full network flow + process telemetry on all monitored workloads	Direct
AU	AU-3	Audit Record Content	Ensure audit records contain sufficient	5-tuple flows + process name + user context + timestamp + policy action	Direct

			information		
AU	AU-6	Audit Record Review	Review and analyze system audit records	Alert export to SIEM; anomaly detection for automated review	Direct
AU	AU-12	Audit Record Generation	Provide audit record generation capability	Continuous logging; configurable retention (recommend 3 years for federal)	Direct
CM	CM-2	Baseline Configuration	Develop and maintain baseline configurations	ADM establishes approved communication baseline per application cluster	Direct
CM	CM-7	Least Functionality	Configure the system to provide only essential capabilities	Block non-approved services via port-level policy enforcement	Direct
CM	CM-8	System Inventory	Develop and maintain an inventory of system components	CSW inventory provides real-time workload asset catalog across hybrid environments	Direct
IR	IR-4	Incident Handling	Implement an incident handling capability	Forensic telemetry enables full incident reconstruction; process + network context	Direct
IR	IR-5	Incident Monitoring	Track and document security incidents	Real-time anomaly alerts; policy violation events; SIEM integration	Direct
RA	RA-5	Vulnerability Monitoring	Monitor and scan for vulnerabilities in the system	Continuous CVE scanning with CVSS scoring on all workloads	Direct
SC	SC-7	Boundary Protection	Monitor and control communications at system boundaries	Micro-segmentation enforces east-west boundaries; ADM validates boundary integrity	Direct
SC	SC-8	Transmission Confidentiality	Implement cryptographic mechanisms to protect transmissions	Detect and block unencrypted sensitive data flows; enforce encrypted protocols	Direct
SI	SI-2	Flaw Remediation	Identify, report, and correct information system flaws	Continuous vulnerability scan; prioritized remediation + compensating controls	Direct
SI	SI-4	System Monitoring	Monitor the system to detect attacks and unauthorized access	Full telemetry with anomaly detection; behavioral baseline deviation alerting	Direct

NIST Impact Level Configuration Guide

Impact Level	CSW Configuration	Telemetry Retention	Enforcement Mode	
Low	ADM + monitoring; simulation policies	6 months	Simulation (alert-only)	Supporting
Moderate	ADM + enforcement on sensitive scopes; full alerting	1 year	Enforcement on sensitive scopes	Direct
High	ADM + full enforcement; process-level monitoring; SIEM integration	3 years	Full enforcement all scopes	Direct

Federal Evidence Package

Evidence Item	CSW Source	Control Family	Frequency
Policy enforcement log	Defend → Policy Workspaces	AC	Per audit
Network flow audit log	Investigate → Flow Search	AU	Continuous
Vulnerability report	Investigate → Vulnerability	RA, SI	Weekly
ADM baseline map	Investigate → ADM	CM	Quarterly
Alert/incident log	Alerts → Dashboard	IR	Monthly
Workload inventory	Manage → Inventory	CM	Monthly
Process anomaly report	Investigate → Process Search	SI	On-demand
Encryption compliance report	Flow Search (protocol filter)	SC	Quarterly